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RESEARCH ARTICLE

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A novel method for determination of the filling level in the feed frame of a rotary tablet press

K. P. Dühlmeyer[a], H. Özcoban[b] and C. S. Leopold[a]

[a]Division of Pharmaceutical Technology, Department of Chemistry, University of Hamburg, Hamburg, Germany; [b]Fette Compacting GmbH, Schwarzenbek, Germany

ABSTRACT

Aim: The aim of this study was to investigate whether the filling level within the feed frame of a rotary tablet press can be quantified by laser triangulation combined with the angle recognition of one paddle wheel via rotary encoder.

Significance: Rotary tablet press feed frames are supposed to assure a uniform die filling and, thus, to guarantee the weight and content uniformity of the resulting tablets. Therefore, a constant bulk availability and flow within the feed frame is crucial and has to be ensured by the feed frame design and the operating conditions. So far, there is no instrument available to monitor the bulk filling level or the bulk distribution within feed frames.

Methods: Calcium phosphate dihydrate was used as model powder. The powder surface level was determined via laser triangulation and the angle position of the paddle wheel was monitored via incremental rotary encoder. The data of both parameters was acquired synchronously and evaluated by in-house written software.

Results: Different powder masses led to significantly different filling level signals. The experiments showed a high reproducibility of the determined filling levels. Furthermore, an influence of the rotational speed on the powder distribution was observed.

Conclusions: The developed instrument may be used for quantification of the volumetric filling level within rotary tablet press feed frames. It may either be used to better understand the powder behavior within feed frames or for improvement of the die filling process by implementing the device into a feedback loop.

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KEYWORDS
Rotary tablet press; feed frame; filling level; process analytical technology; laser triangulation; paddle wheel

Introduction

With an estimated market value of more than $300 billion per year approximately 80% of the total production of pharmaceuticals is tablets [1]. Drug stability, patient compliance, and high production rates are just a few examples for the preferred application of tablets when it comes to the medical care of the population. High quality requirements on pharmaceutical products and financial risks caused by out of specification production has led to an increased interest in PAT tools by the pharmaceutical industry to secure production processes by the early recognition of potential problems [2].

Modern pharmaceutical rotary tablet presses produce up to 1.6 million tablets per hour [3]. During tableting, the consistent and uniform filling of the dies is often the production rate-limiting step during compression of powders or granules with rotary tablet presses [4]. These deviations in die filling may lead to variations in the content of the active pharmaceutical ingredient in the tablets, which represent a major problem in the industry. Therefore, it is of great importance to better understand the die filling process to ensure a constant tablet quality. One key factor of the die filling process is the bulk availability, that is, the quantity of bulk material that is provided by the feed frame within the die filling area [5]. The extent of die filling is dependent on the bulk availability and the bulk condition, that is, the flow of the bulk material within the die filling area of the tablet press [5–8]. As feed frames are dynamic systems because of the rotation of the paddle wheels, forces which act on the bulk material within the chambers are varying. These forces lead to variations in the pre-compaction, that is, the bulk density of the powders and thus the flow behavior into the dies. The feed frame's performances depends primarily on the flow properties of the bulk material which result from different parameters, such as particle size, size distribution, shape, packing, moisture, and cohesion of the powders [7].

In this study a 3-chamber filling system was investigated substitutional for the wide variety of feed frames (Figure 1). Feed frames exist in different designs with regard to the number, the dimensions and shapes of the paddles, the number and arrangement of the chambers, and their heights. The powder enters the feed frame through an orifice at the top and paddles force the powder to exit through a slit at the bottom. Because of stress and strain applied to the bulk within the feed frame there are several effects showing the complexity of powder flow phenomena through the feed frames and the impact of the feed frames on the powders [9–12]. It is well known to process design scientists that the operating parameters, such as the size of die cavity (diameter, depth, and geometry), the distance between the dies,

CONTACT C. S. Leopold claudia.leopold@uni-hamburg.de Division of Pharmaceutical Technology, Department of Chemistry, University of Hamburg, Bundesstr 45, 20146 Hamburg, Germany

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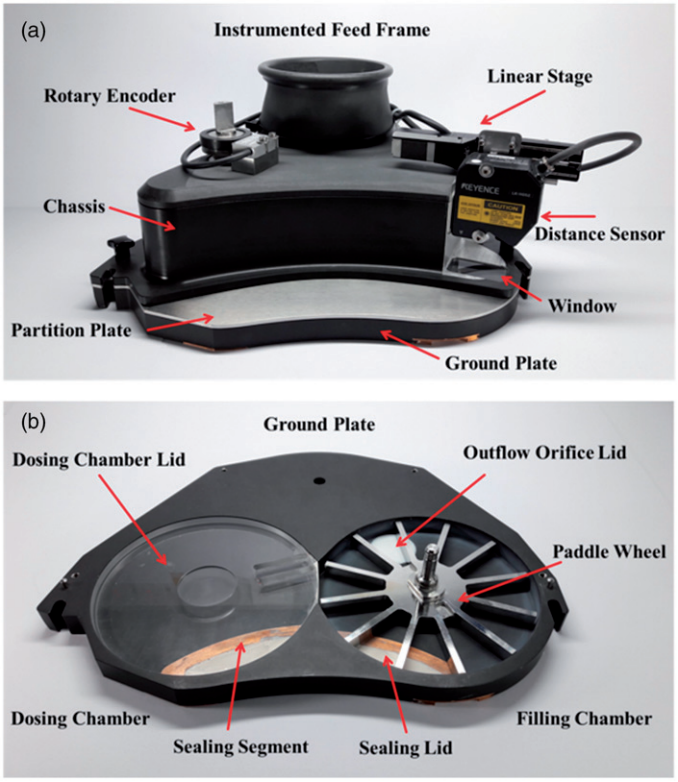


Figure 1. Instrumented feed frame (a) and ground plate of the feed frame with different lids (b).

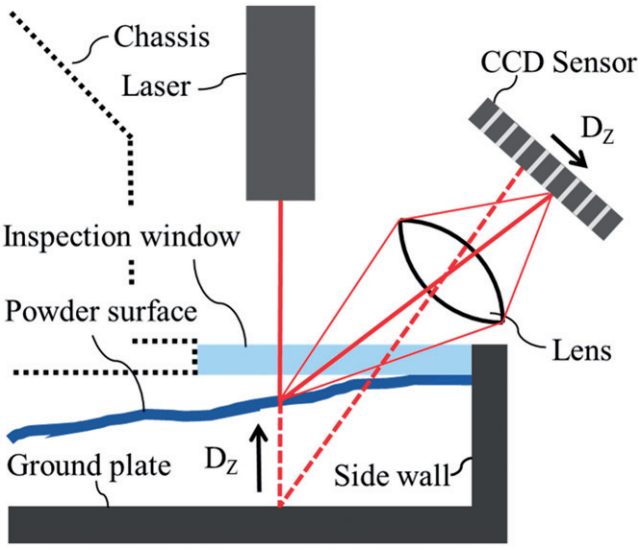


Figure 2. Schematic illustration of the cross-section of the experimental setup at the measuring position of the powder surface by the laser triangulator within the filling chamber of the feed frame. The laser beam (red solid line) enters the filling chamber vertically and is diffusely reflected at the powder surface. The red dashed line shows the beam path of the laser in an empty chamber. The vertical position displacement of the laser spot (DZ) is monitored laterally by a CCD video chip after the diffuse reflection has been focused by a lens. The refraction of the optical beam during passing through the inspection window of the filling chamber was omitted in the illustration.

the die disc turret speed, the shape, number, geometry, and speed of the paddles in the feed frame can all have impact on the tablet weight uniformity and, therefore, also other tablet properties [4,9,10,13–18]. In rotary tablet presses the compression force is directly related to the degree of die filling because the compression rolls and therefore the distance between the punches in the compression unit are generally fixed during the ongoing production process [19]. Thus, several of the resulting properties of the tablet are affected by the degree of die filling including density, mechanical strength, size (thickness), porosity, and the amount of elastic stress stored in the compact which all may influence the drug release characteristics [4].

In this context, it is obvious that the uniformity of die filling by the feed frame shows a high influence on the uniformity of the resulting tablets. Several research studies have already been performed on the investigation of powder phenomena inside feed frames by DEM simulations [20–23]. However, there is no device available yet for monitoring the distribution of powders inside the feed frame in reality. One study shows a correlation between the chamber's filling level and a baseline shift of the spectra obtained by NIR probes [24]. Baseline shifts however, are complex events and they are susceptible for disturbances if used as a measure for accurate distance determinations. For example if the bulk material is exchanged, bands may occur at the wavenumber which was selected for the distance calculation. In this case, another wavenumber has to be selected which might possibly result in a different correlation between the baseline shift and the filling level. Therefore, a more universal technique is desirable to monitor the powder behavior inside the feed frame.

Optical triangulation sensors have been commonly used in the industry for many decades. Briefly, a beam projected from a laser produces a light spot on a diffuse surface. A portion of the light is scattered from the surface and imaged on a detector by a converging lens. If the surface is moved such that the light spot is shifted along the path of the laser beam, the image of the light spot on the detector is also shifted. The displacement of the image on the detector can be used to determine the displacement of the surface [25]. Even though the dusty environment inside a feed frame is challenging for optical distance measurements, a successful implementation of optical triangulation sensors even under these harsh conditions appears possible (Figure 2) [26].

Materials and methods

Materials

Dibasic calcium phosphate dihydrate (Emcompress®, JRS Pharma, Budenheim, Germany) further referred to as DCP was selected as model powder for all experiments. Because of its common use as excipient in tableting and its excellent flow properties [27], DCP is well suited for the presented study. Although the introduced device is of particular interest for the monitoring of poorly flowing materials, an excellent flowing powder was selected for this proof of concept study to minimize interfering effects of the powder regarding the repeatability of the measurements.

Methods

Technical equipment

Tablet press feed frame. In the present study a rotary tablet press (Fette 102i, Fette Compacting, Schwarzenbek, Germany) served as a power unit to drive the paddle wheel in the feed frame. The novel instrument is based on a 3-chamber feed frame (Fill-O-Matic®, Fette Compacting, Schwarzenbek Germany) and is illustrated in Figure 1(a).

The experimental setup was simplified by isolating the filling chamber from the other parts of the filling system, through tight

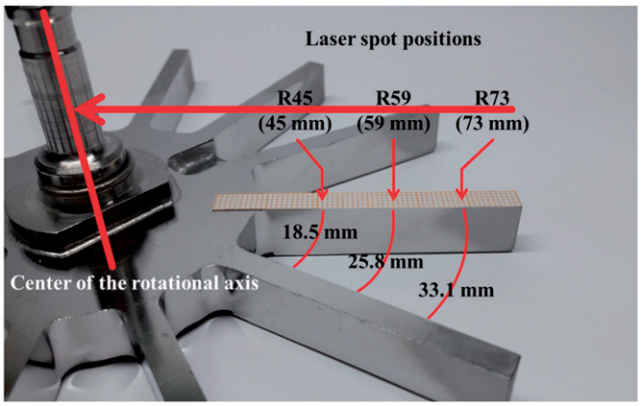


Figure 3. Paddle wheel with the stage positions for monitoring the powder surface.

coverage and thus deactivation of the dosing chamber, the sharing chamber (located above the partition plate), the outflow orifice, and the sealing segment with lids made of different materials. In the present study a simple beam shaped paddle wheel was selected (Figure 1(b)).

Laser triangulator. For measurement of the powder surface level a laser triangulator, that is, an ultra-high speed and high accuracy laser displacement sensor was chosen (LK-H052, Keyence, Osaka, Japan), controlled by a main controller unit (LK-G5001P, Keyence, Osaka, Japan). Technical specifications for both may be found in the respective data sheets. The location of the laser triangulator was chosen to be placed as close as possible to the die filling area.

Sensor-internal parameters for determination of the displacement signal were identified by preliminary experiments. Settings which generate minimum noise of the displacement signal under extreme measuring conditions with a sufficiently high signal frequency were selected.

Linear precision stage. The displacement sensor was mounted to a linear precision stage (LPTM-30-50, OWIS, Staufen, Germany) to enable investigations of the powder behavior at different pitch circles inside the filling chamber. A step motor moves the stage within a range of 28 mm on a linear axis at a speed of 4 mm/s. The speed of the linear stage on which the laser triangulator is mounted affects the time interval between two measurements at different positions within the filling chamber, as these positions have to be approached sequentially. However, monitoring of the powder surface at a single position in the chamber during production is probably sufficient for determination of the uniformity of the powder distribution within the chamber, so that the sensor position remains constant during production.

Deviations of the ideal alignment of the linear stage (Euler angles) and, therefore, the displacement sensor to paddle wheel rotational axis were determined by measurements of paddle wheel- and ground plate signals in an empty feed frame. These deviations were taken into account in the evaluation software and thereby compensated.

The distances between the stage positions, that is, the positions for monitoring the powder surface and the center of the paddle wheel rotational axis were selected to be 45, 59, and 73 mm.

The lowest level of the stage position (45 mm), referred to as R45 throughout the article is the deepest possible position for measurements inside the filling chamber, limited by the feed frame's chassis and the sensor's size. The highest level of the stage position, referred to as R73 is located 28 mm farther outside from R45. Up to R73 a triangulation to the ground plate is possible. Beyond R73, the laser spot on the ground plate is hidden by the feed frame's side wall. Thus, measurements beyond R73 are only possible if the feed frame is filled with powder. Because of the circular geometry of the filling chamber, the powder surface profiles obtained at different radiuses correspond to different circular arcs, that is, distances inside the paddle wheel interspaces. In this study, the surface profiles of the powder obtained at R45, R59, and R73 correspond to 18.5, 28.8, and 33.1 mm of distance between the paddles (Figure 3).

Incremental rotary encoder. For monitoring the angular position of the paddle wheel a bearingless magnetic incremental rotary encoder was mounted (Figure 1(a)). The system consists of a magnetic ring and a sensor head (8.RI50.048.2000.112 and 8.LI50.1111.1064, Fritz Kübler, Villingen-Schwenningen, Germany). Further technical specifications of both, the magnetic ring and the sensor head may be found in the respective data sheets.

Data acquisition hardware. The hardware used for data acquisition consists of a high speed backplane (NI cDAQ-9174, National Instruments, Austin, USA) equipped with an analog input module (NI-9215, National Instruments, Austin, USA) and a digital input module (NI-9401, National Instruments, Austin, USA). Further technical specifications may again be found in the respective data sheets.

Data acquisition software. The data acquisition and the adjustment of the stage position was performed by an in-house written software script (LabVIEW, National Instruments, Austin, USA). The software allows to acquire distance values of the displacement sensor, the stage position, and the angle position of the encoder. The sampling tact and rate of these tasks which acquire these values are concerted and synchronized. All channels apart from the stage position were stored in a 1D-array data format inside a Technical Data Management System file, referred to as TDMS file at a frequency of 2 kHz. The data of the stage position was stored only once per s to prevent an enlargement of the raw data files.

Data evaluation software. The data evaluation was performed by an in-house written software script (LabVIEW, National Instruments, Austin, USA). The software evaluates defined parts of the raw data TDMS files and stores the results and evaluation setting information in a newly created TDMS file. Because of the synchronized acquisition of sensor data, the arrays of both the encoder and the displacement sensor share a collective index. Thus, the index found in the encoder array which is related to the beginning of a paddle wheel revolution can be transmitted to the distance array and therefore used to read out the corresponding data of the displacement sensor. The data is split into subsets which correspond to complete paddle wheel revolutions. Sections of the data which contain the paddle wheel signals were excluded from further processing, to prevent an influence of the paddles on the powder filling level calculation. The section sizes were determined by measurements in a powder-free system. As a result, the initial data was split in 12 subsets belonging to the 12 paddle wheel interspaces. Subsequently a reduction of powder dust artifacts is carried out. If the powder is shifted across the paddles, artifacts may occur in the signal behind the preceding paddle resulting in a shift of the filling level to higher values. This

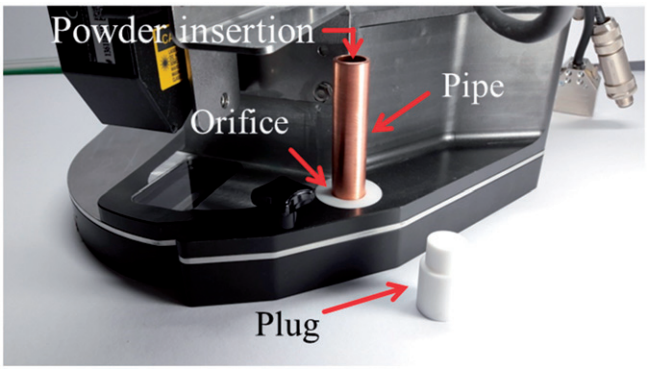


Figure 4. Location of powder insertion.

phenomenon was reduced by an automatic exchange of the artifact signals by the respective minimum value measured inside the paddle wheel interspace. Finally the remaining data is reduced to a mean value which represents the mean filling level over all paddle wheel interspaces at the respective measurement radius, further referred to as the filling level. The mean powder profiles, the mean filling levels of the paddle wheel interspaces, the filling level, and the corresponding standard deviations were streamed into an initialized TDMS evaluation file.

Powder characterization and powder loading into the feed frame

The particle size distribution of the investigated powder before and after each experiment were determined via laser diffraction (Helos®/Rodos®, Sympatec, Clausthal-Zellerfeld, Germany). Compressed air of 1.5 bar was used to disperse the powder. All experiments were performed in triplicate. Bulk and tapped densities of the investigated powder before and after each experiment were determined in a 250 mL graduated cylinder, mounted on a tapping machine (STAV 2003J, Engelsmann, Ludwigshafen am Rhein, Germany). About 100 g of powder were used to determine the bulk volume (V0) as well as the tapped volume after 1250 taps (V1250). Each sample was measured in triplicate. In the flow through an orifice method, the time required for the powder (100 g) to flow through a stainless-steel funnel with a 10 mm orifice was measured using a powder flowability tester (BEP2, Copley Scientific, Nottingham, United Kingdom) before and after each experiment. Each sample was measured in triplicate.

To make sure that the amount of powder is inserted quantitatively into the filling-chamber, the drill-hole of the standard inspection window was used for filling (Figure 4). Therefore, a closable orifice was implemented. The powder was inserted by a removable pipe.

To achieve a sufficient uniformity of powder distribution, 7 g of DCP were first inserted into the chamber and the paddle wheel was rotated counterclockwise for 720°. Subsequently, another 7 g of DCP were inserted and the paddle wheel was rotated clockwise again for 720° such that the space between paddle wheel and ground plate is mostly filled with powder. Finally, the interspaces between the paddles were filled with equal portions of the remaining powder. Only into the first interspace 2 more grams of powder were inserted compared with the remaining interspaces, because during the rotation of the wheel (in 30° steps) a part of the powder will redistribute leading to a decrease of the filling level inside the first interspaces. This procedure leads to an acceptable uniformity of powder distribution and is sufficient for the current study. To ensure that the correct total amount of DCP is loaded into the filling chamber, the powder was collected, and weighed again after each experiment.

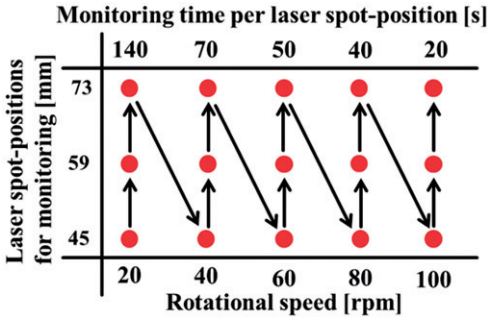


Figure 5. Illustration of the experimental sequence. The points represent the respective amount of DCP powder (80, 100, 120, 140, or 160 g).

Design of experiments

A factorial-lattice design with 75 experiments, each determined in triplicate was used to investigate the filling level and powder phenomena taking place inside the feed frame. The influence of three factors, namely the "powder mass" inside the chamber, the "rotational speed" of the paddle wheel, and the "laser spot position" for monitoring the powder surface were investigated. Both, the "powder mass" and the "rotational speed" were varied at five levels whereas the "laser spot position" was varied only at three levels. These levels have been selected by preliminary experiments.

The experimental runs that included an identical amount of powder were carried out in a standardized sequence without refilling the chamber for each experimental run again with powder except for the repeat experiments (Figure 5). The sequence started at the lowest level of rotational speed, subsequently the paddle wheel was accelerated to next higher level until the highest level of the rotational speed was reached. On each level of the rotational speed the laser was shifted along all positions for monitoring before the paddle wheel was accelerated to the next higher level. The experimental design was planned and statistically evaluated using the Design Expert® software (Stat-Ease, Minneapolis, USA). The design matrix evaluation for the response surface quadratic model resulted in a variance inflation factor of 1.0 for all terms of the model indicating that coefficients are well estimated even if multicollinearities occur. Additionally, the experimental design provides high degrees of freedom for lack of fit and pure error which ensures a valid lack of fit test. The entire design matrix evaluation for the used experimental design can be found in the Supplementary data section.

Results and discussion

Evaluation of the powder characteristics

The powder volumes which were determined from the bulk and tapped densities as well as the average particle size of the investigated powder samples did not change significantly over the course of the experiments. However, slight changes in the powder flow were observed. In Table 1 the results of the measurements of the powder properties of untreated DCP (DCP_{before}) and DCP taken from the filling chamber after the applied shear stress by the experimental sequence (DCP_{after}) are shown. These results allow the conclusion that the different degrees of shear stress applied to the powder before the powder surface profiles were recorded can be neglected in this study.

Table 1. Powder characteristics before and after the recording of the powder surface profiles.

Powder properties	DCP_{before}	DCP_{after}
Particle size d50/d90 (µm)	220.75 (d50)/347.17 (d90)	221.77 (d50)/349.52 (d90)
	221.28 (d50)/346.69 (d90)	225.47 (d50)/348.14 (d90)
	223.4 (d50)/347.79 (d90)	226.37 (d50)/354.48 (d90)
Powder volumes	110/108/106/96/94/92	110/108/106/96/94/92
V0/V1/V10/V500/V1250/V2500 [ml/100g]	110/108/106/96/94/92	110/108/106/96/94/92
(determined by the bulk and tapped densities)	110/108/106/96/94/92	110/108/106/96/94/92
Powder flow through an orifice (s/100 g)	8.13 ± 0.05	8.6 ± 0.1

All experiments were carried out in triplicate.

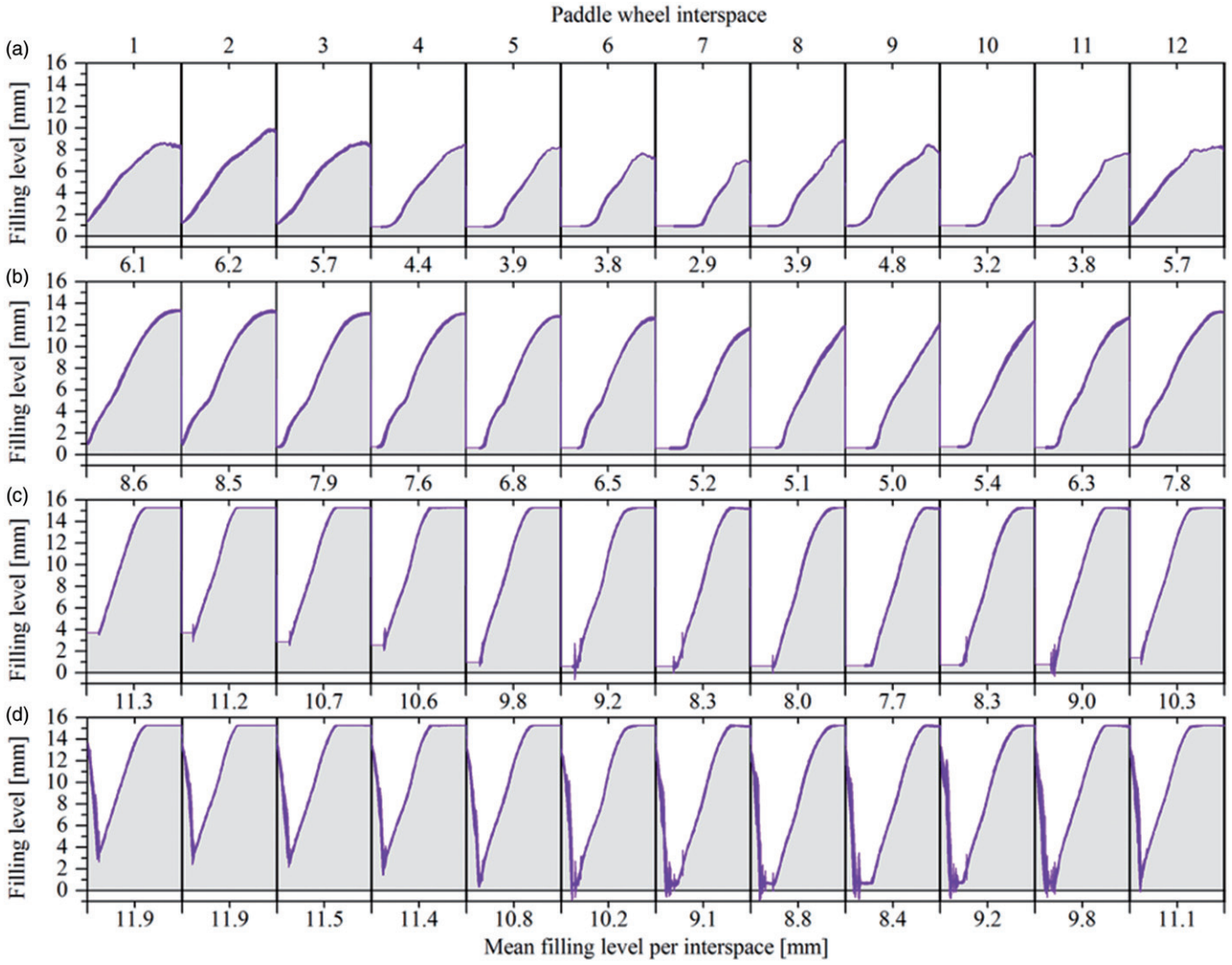


Figure 6. Mean powder surface profiles ($n = 30$) of 120 g DCP at 60 rpm rotational speed obtained at the laser spot positions: (a) R45, (b) R59, (c) R73, and (d) R73 without artifact filter.

Evaluation of the mean powder profiles

Throughout the evaluation process 225 mean powder surface profiles were created each out of 30 subsequent paddle wheel rotations. In Figure 6 three mean profiles exemplary for the monitored laser spot positions at the center levels of the "powder mass" and the "rotational speed" are displayed. The profile that was obtained at R73 is shown twice to illustrate the influence of the artifact filter which was implemented inside the algorithm. The standard deviations of the layered profiles are displayed as bands above and below the mean. The mean filling levels calculated separately for each of the twelve paddle wheel interspaces are displayed at the bottom of the respective interspace's profile. Results show a powder accumulation in front of the pushing paddles. Occasionally, a preceding powder avalanche was observed in front of a main avalanche.

An increase of the filling level in the direction from R45 to R73 was observed at equal levels of the "powder mass" and "rotational speed." This increase of the filling level is probably reflecting centrifugal and shear forces caused by the rotational movement. In the present study no investigations were performed regarding a relationship between the profile shape and flowability of the powder. However, different flow properties might be reflected in the surface profile measured in the feed frame. The link between the powder characteristics, the powder distribution within the feed frame and the die filling process is among the most important relationships in context with the die filling process. However, in

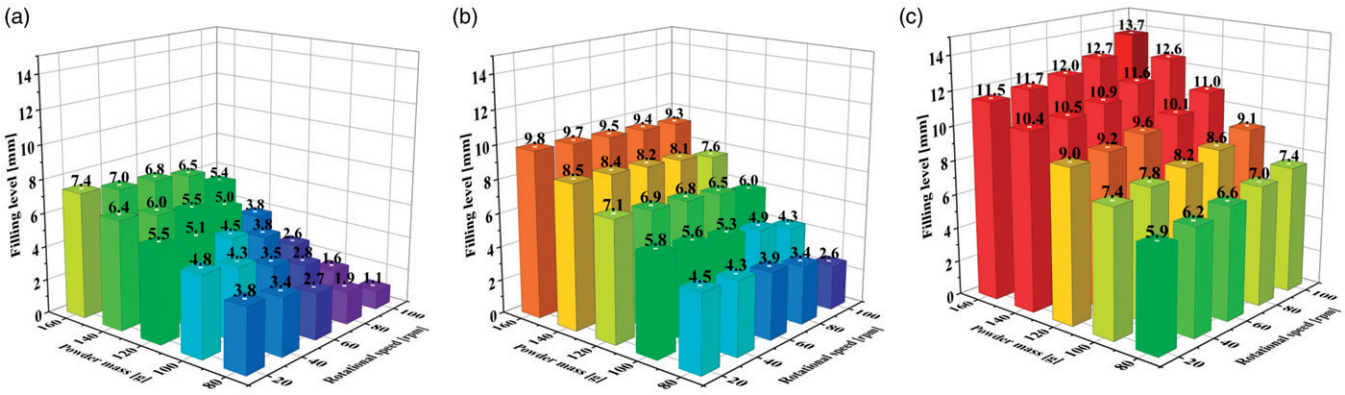


Figure 7. Filling levels at the three laser spot positions for monitoring (a) R45, (b), R59, and (c) R73.

the present article the focus was on the development of the instrumented feed frame. This relationship might be investigated in further studies.

Significant differences between the filling levels of the paddle wheel interspaces suggest that the distribution of the powder inside the chamber is inhomogeneous. This is probably a result of the chamber's filling procedure used in this study. Transferred to a tableting process, the information on the inhomogeneity of the powder distribution inside the feed frame may be used to determine variations in powder availability for die filling which might cause tablet mass deviations.

Because of the small standard deviations of the powder profiles, it can be assumed that the redistribution of the powder volume between the interspaces occurs slowly at least within the 30 revolutions of the paddle wheel. The consistency of the powder profiles may be explained by a principal shift of the powder and therefore a stay in an interspace. However, there are also indications of a slight stream of powder from the previous into the following interspace. At R73 high deviations of the signal behind the paddles indicate that the mentioned redistribution effects indeed occur. These artifacts probably result from powder shifts above the paddle and are intensified with increasing rotational speed. This intensification is probably caused by mass inertia effects which become stronger at higher rotational speed levels because of an increased powder acceleration. This hypothesis is supported by DEM simulations [21]. The powder which is shifted above the paddle remains at higher trajectories and therefore pretends a powder surface which is detected by the displacement sensor. Additionally, the duration of the measurements decrease if the paddle wheel is accelerated. The impact of these dust artifacts on the determination of the filling level was reduced by the software filter described in the method section.

Overall, the radiuses R45 and R59 show lower powder levels, less dust artifacts, and lower orbital speeds than the radius R73 which all support the smaller radiuses to be more suitable for a calculation of the true filling levels, particularly regarding a distortion by dust artifacts. Comparing evaluations with and without filtering out of dust artifacts, the calculated filling levels obtained at R45 and R59 are very similar. However, at R73 a strong reduction of the filling level by reducing the dust artifacts was observed. Because of the volumetric measurement method, no information on the powder density inside the system is obtained.

No investigations were performed on the penetration depth of the laser beam into the powder surface. Presumably, different powders or powder densities lead to different immersion depths or spot sizes of the diffuse reflected light. Both can influence the distance signal.

A challenge for a measurement by laser triangulation might be the blinding of the inspection window. Adhesive products may lead to adhering powder layers to the window surface and thus to turbidity of window. Moreover, a static charge of the inspection window by itself could cause adhesion of particles, in the worst case inside the optical path of the laser beam. However, in this study DCP did not impair measurements noticeably by blinding of the inspection window.

Evaluation of the filling levels

A quadratic model was selected to describe the effect of the investigated factors on the filling level. According to the analysis of variance (ANOVA) the model explains 98.96% of the relationship between the factors and the response ($R^2 = 0.9896$). The F value of 3049 implies that the model is significant. There is only a 0.01% probability that such a high F value results from noise. However, there is still a significant lack of fit (LOF =321.62), indicating that some differences between the actual and predicted values do not result from noise; instead, there are other reasons why this might occur. In general, a wide range of factor levels but also unconsidered effects or cofactors may lead to a lack of fit inside a modeling approach. Here, the cause might be the restriction of the increase in the filling level by the chamber's ceiling. If the powder reaches the chamber's ceiling or inspection window's surface, a further increase of the filling level is impossible.

Despite the significant LOF the predicted R^2 of 0.9890 is in reasonable agreement with the adjusted R^2 of 0.9896 (difference <0.2) and an adequate precision ratio of 233.91 (>4 is desirable) indicate that the model can be used to navigate the design space. According to the ANOVA all terms of the model significantly influence the filling level (p values $<.05$). The factors "powder mass" and "rotational speed" show a linear influence on the filling level. All factors interact with each other; the "laser spot position" (R45, R59, R73) is the only quadratic term within the model.

Generally, the calculated filling levels show expected trends. An increase of the powder mass inside the system leads to an increase of the filling levels at all radiuses. The highest filling levels were always observed at R73 regarding equal levels of the factors "powder mass" and "rotational speed." The rotational speed affects the filling level in different directions dependent on the "laser spot position." An acceleration of the paddle wheel results in a decrease of the filling level at R45 and R59 but in an increase at R73. As already mentioned, the powder is shifted outward by the rotational movement of the beam-shaped paddles and accumulates at the chamber's side wall. The standard deviation of the filling level was maximal 0.11 mm over all experiments and in

average 0.02 mm which indicates a high reproducibility. In Figure 7 the filling levels at the three different laser spot positions are shown.

From the instrumental developer's point of view, the knowledge about the general filling conditions is necessary to find an optimal or universal spot for monitoring of different production conditions. In the edge region of the chamber a high filling level is present at most process conditions which might lead to problems with regard to an accurate determination of usable filling level values.

If further studies lead to the conclusion that an independent variation of the paddle wheel speed and the filling level turns out to be of advantage, it might be imaginable to build up a feedback loop to a valve above the feeder that adjusts the powder inflow. Thereby, the powder acceleration within the feeder may be varied independent of the filling level. Whether or not such a feedback mechanism is of advantage has to be discussed elsewhere.

Conclusions

Laser triangulation combined with an angle recognition of the paddle wheel via rotary encoder may be used for monitoring of the volumetric filling level within rotary tablet press feed frames. This method is suitable for both, the characterization of the uniformity of the powder filling level between the individual paddle wheel interspaces as well as the rotational speed-dependent behavior of powders within the chamber. The resulting data might provide useful information for optimization of the paddle wheel speed and/or for selection of the paddle wheel design depending on the powder behavior within the chamber's. Investigations on the inhomogeneity of the powder distribution might also be useful to better understand tablet mass deviations during production processes. To investigate the expected correlations between the filling level within the feed frame and the extent of die filling, the powder profiles and compression forces may be recorded simultaneously to correlate the powder profiles related to filling processes of individual dies with the corresponding compression forces. Therefore, further investigations will focus on online monitoring of the filling level during tableting.

The device introduced in this article, may be a useful tool to determine a pre-run time of the feed frame before starting with the tablet production. However, blinding of the inspection window can be challenging with this device. Thus, there is a need to optimize the window surface in the future for example by choosing different window materials, surfaces, or tools against blinding.

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Disclosure statement

No potential conflict of interest was reported by the authors.

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